

Bandstructure engineering

7

7.1 Introduction

The idea of making artificial solids with bandstructures tailored to particular applications has long fascinated condensed matter physicists. In this chapter, we shall explore some of the ways in which bands with desired properties can be engineered using what has been termed 'chemical architecture'. A very simple example of this is the use of semiconductor alloys, in which a wide-gap semiconductor and a narrow-gap semiconductor are combined to give a substance with a desired intermediate band-gap. A second example is the semiconductor superlattice or heterostructure; here, very thin layers of different semiconductors are superimposed. A third approach is to use small organic or inorganic molecular units to build up solids with desirable metallic, semiconducting or even superconducting properties. We shall treat each of these cases in turn, starting with the semiconductor systems.

7.2 Semiconductor alloys

The previous chapter mentioned the fact that direct-gap semiconductors are used in opto-electronic applications. Many of these applications involve the emission of light by recombination of an electron and a hole across the band-gap, or absorption of light of the band-gap energy to create an electron-hole pair and hence produce some form of photoconduction;¹ The size of the band-gap is therefore of great importance.

Figure 7.1 shows the band-gaps and lattice parameters for some of the more common elemental and binary semiconductors. The band-gaps available are not generally optimised for practical devices. A desire to improve on this has led to the field of bandstructure engineering, where a variety of artificial structures are used to provide bandstructure optimised to a particular application.

The simplest bandstructure engineering involves making ternary alloys such as (Ga,Al)As and (Hg,Cd)Te in order to achieve a desired band gap. I list three examples of the uses of such alloys.

$\text{Ga}_{1-x}\text{Al}_x\text{As}$ is technologically important because GaAs and AlAs form a solid solution over the entire ($0 \leq x \leq 1$) composition range with very little variation of lattice parameter (see Fig. 7.1); this means that multilayers of very high quality can be fabricated (see below). A variable direct band-gap is obtained for the range $0 \leq x \leq \sim 0.35$; for $x \geq \sim 0.35$ the band-gap is indirect.

7.1	Introduction	65
7.2	Semiconductor alloys	65
7.3	Artificial structures	66
7.4	Band engineering using organic molecules	75
7.5	Layered conducting oxides	78
7.6	The Peierls transition	81

¹ See e.g. *Semiconductor physics*, by K. Seeger (Springer, Berlin, 1991), Chapters 12 and 13; *Optical properties of solids* by M. Fox, (Oxford University Press, Oxford, 2001), *Low-dimensional semiconductor structures*, by M.J. Kelly (Clarendon Press, Oxford, 1995), Chapter 18.

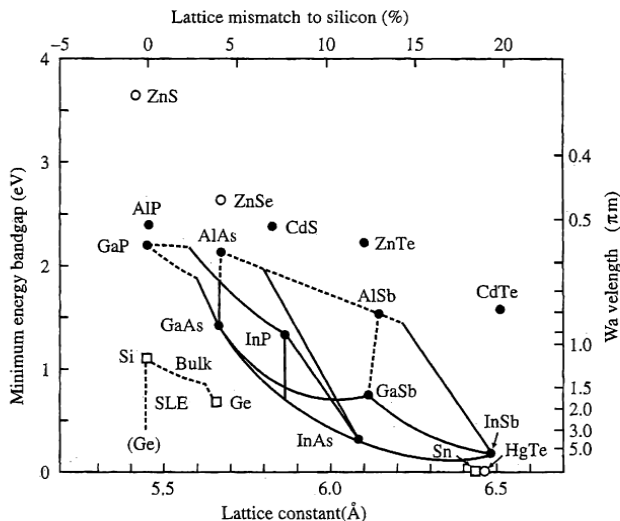


Fig. 7.1 Band-gap versus lattice parameter for some of the more common semiconductors. The curves indicate commonly-used alloys such as (Ga,Al)As, (Ga,In)As etc.; solid curves represent direct gaps and dashed curves indirect gaps.

The band-gap of $\text{Ga}_{1-x}\text{In}_x\text{As}$ may be adjusted over an energy range which coincides with the low-attenuation region of many optical fibres.

$\text{Hg}_{1-x}\text{Cd}_x\text{Te}$ continues to be of great importance in the fabrication of infrared detectors covering the $10\text{ }\mu\text{m}$ ($x = 0.2$; $E_g \approx 100\text{ meV}$) and $5\text{ }\mu\text{m}$ ($x = 0.3$; $E_g \approx 200\text{ meV}$) atmospheric windows (regions where the atmosphere has little absorption). The $10\text{ }\mu\text{m}$ region also contains the peak thermal emission of $\sim 300\text{ K}$ things, including human beings, so that there is a variety of medical (thermal imaging), meteorological and more sinister applications. Exhausts and jet engines emit well in the $5\text{ }\mu\text{m}$ range, and so the applications of $\text{Hg}_{1-x}\text{Cd}_x\text{Te}$ with $x = 0.3$ can be easily imagined.

7.3 Artificial structures

We now turn to a variety of structures which can be grown using techniques developed over the past twenty years or so.

7.3.1 Growth of semiconductor multilayers

The growth of high-quality semiconductor multilayers is described as *epitaxial* (the word is derived from the Greek words *e*pi (upon) and *taxis* (arrangement)). The implication is that layers grow on a suitable single-crystal substrate, continuing the crystal structure of that substrate. The layers are thus supposed to be crystallographically well ordered. Two techniques are commonly used to grow epitaxial layers of semiconductors, Molecular beam epitaxy (MBE) and Metal-organic vapour-phase epitaxy (MOVPE; also known as OMVPE, MOCVD and OMCVD).

- **MBE** Figure 7.2 shows the main components of an MBE machine. The elements which make up the semiconductors to be grown evaporate from

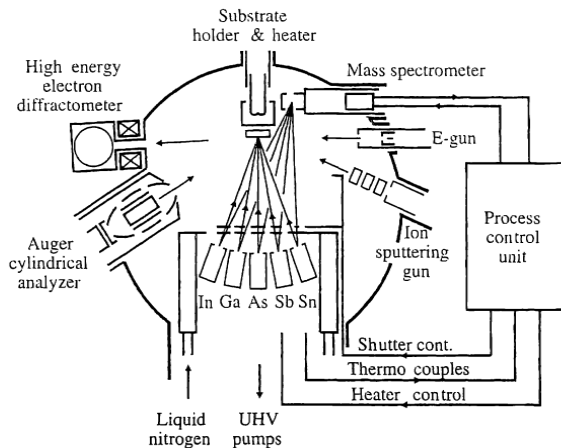


Fig. 7.2 Schematic of an MBE machine.

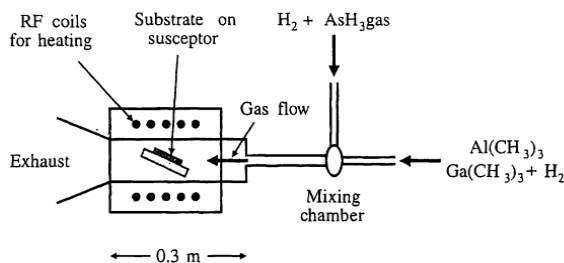
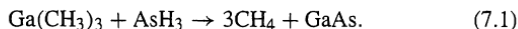


Fig. 7.3 Schematic of an MOVPE machine.

Knudsen cells at a rate controlled by the cell temperature; cells are also provided for dopants. The evaporated atoms form a beam which travels towards the substrate. Typically the evaporation rates are such that about one monolayer is deposited per second. The composition of the layers is controlled by shutters which can swing in front of each cell in about 50 ms, cutting off the beam from that cell. The substrate is kept at a well-defined temperature to ensure that the deposited atoms are reasonably mobile, so that they spread out over the substrate in monolayers rather than forming clusters. The whole chamber must be very well evacuated to prevent spurious dopants entering the layers; typical pressures in an MBE machine are $\sim 10^{-11}$ mbar. A number of *in situ* diagnostic techniques to monitor the growth of the multilayer are provided.

- **MOVPE** Figure 7.3 shows a schematic of an MOVPE machine. The components of the semiconductor to be grown travel to the substrate as gaseous precursors formed by reacting the elements with organic radicals; the precursors are carried along in a stream of hydrogen. Close to the substrate, which is heated by a radiofrequency coil, the precursors react, depositing the semiconductor on the substrate. A typical reaction is



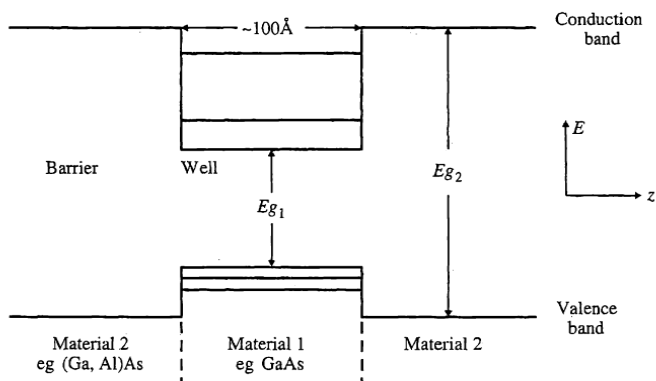


Fig. 7.4 Band-edges in a quantum well as a function of distance in the growth direction z . The energies of the subbands within the well are shown schematically.

Very fast solenoid valves enable the sources of the various components to be switched on and off rapidly.

7.3.2 Substrate and buffer layer

The substrate will be a high quality single crystal, usually of one of the materials to be included in the multilayer. Often the substrate will be semi-insulating GaAs (see Section 6.4.8). The first layer to be grown is called the *buffer layer*; it is often the same material as the substrate (but undoped), and is designed to 'smooth out' the lumps and bumps of the latter, to provide an atomically-smooth top surface on which to grow the active parts of the multilayer. More recently, *superlattice buffers* consisting of alternating thin ($\sim 1\text{--}2\text{ nm}$) layers of e.g. GaAs and AlAs have sometimes been used for this purpose.

7.3.3 Quantum wells

The simplest multilayer system or *heterostructure* (*hetero* = more than one; i.e. the structure is made up of more than one semiconductor) is the *quantum well*. A thick (several hundred nanometres) layer of a wider gap material such as (Ga,Al)As is grown (the barrier material), followed by a $\sim 2\text{--}100\text{ nm}$ layer of a narrower gap material such as GaAs (the well), followed by another thick layer of wider gap material. Figure 7.4 shows the resulting conduction and valence-band-edges; the relative heights of the discontinuities in the conduction- and valence-band-edges are intrinsic properties of the two materials involved.

The narrower-gap material forms a one-dimensional potential well in the conduction and valence bands; thus, in the z (growth) direction, the electron and hole levels are bound states of the well, known as subbands. The well will contain three sets of subbands,

- the electron subbands,
- the light hole subbands and
- the heavy hole subbands,

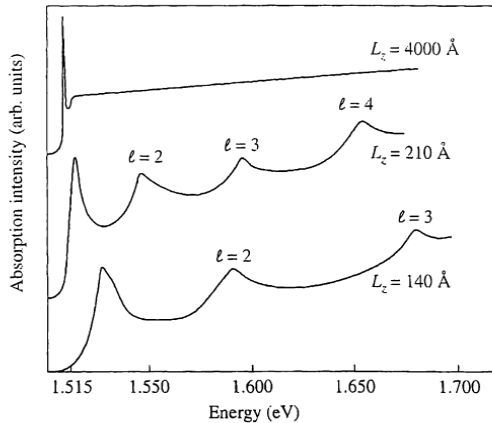


Fig. 7.5 Optical absorption of a series of GaAs-(Ga,Al)As quantum wells of differing widths L_z at 4.2 K. (After R. Dingle *et al.*, *Phys. Rev. Lett.* **33**, 827 (1974).)

each subband within a set being labelled by a quantum number $i = 1, 2, 3, \dots$. Motion in the xy plane will be unrestricted; we therefore have a two-dimensional carrier system in the well.

7.3.4 Optical properties of quantum wells

The interband optical absorption or emission of a quantum well will be caused by transitions between hole and electron subbands. The selection rules are determined by the overlap between the electron and hole wavefunctions. Wavefunctions of the same quantum number (or quantum numbers differing by two) will have similar (spatial) shapes and the transition will be strong; wavefunctions whose quantum numbers differ by one will be of opposite symmetry, and so transitions between them will be weak. In summary, the selection rules are²

- $\Delta i = 0, 2$: strong, allowed transitions;
- $\Delta i = 1$: weak, 'forbidden' transitions.

As there are two sets of hole subbands, strong transitions will occur in pairs, e.g. (i th heavy-hole subband to i th electron subband) and (i th light-hole subband to i th electron subband). Figure 7.5 shows the optical absorption of a set of quantum wells of differing well width. The wells are of rather low quality, so that individual transitions are hard to distinguish. However, it is plain that

- the transition energies increase as the wells get narrower, due to the increased subband confinement energy;
- the absorption increases in steps, reflecting the step-like form of the electron-hole joint density of states in two dimensions;
- excitons are very prominent, because the confinement in the well enhances the exciton binding energy by holding the electron and hole closer together than in the bulk semiconductor.

Figure 7.6 shows the transmission of a set of much higher quality GaAs-(Ga,Al)As quantum wells. Unfortunately the detector used did not give a flat

²For a detailed derivation of these selection rules, see M. Fox, *Optical properties of solids*, (OUP, 2001).

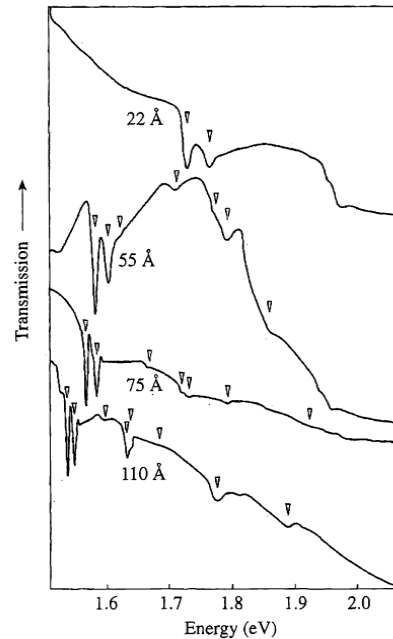


Fig. 7.6 Optical absorption of a series of GaAs-(Ga,Al)As quantum wells of differing widths L_2 at 55 K. (Data from D.C. Rogers *et al*, *Phys. Rev. B* **34**, 4002 (1986).)

response so that the steps in absorption cannot be seen so clearly; however, the pairs of strong transitions can now be resolved. Note that the excitons involving light-holes are almost as strong as excitons involving heavy holes (cf. the bulk case (Fig. 6.8), where the latter dominate); this is because quirks of the valence-band bandstructure give the light-hole subbands quite a large in-plane (xy) effective mass, and therefore a large density of states, comparable to that of the heavy holes.

7.3.5 Use of quantum wells in opto-electronics

There are several reasons for the use of quantum wells in opto-electronic applications, e.g.

- the energy of the fundamental optical transition can be varied by varying the well width;
- the heavy- and light-hole degeneracy at the Brillouin-zone centre is broken, removing complications associated with scattering etc.;
- all of the transitions are excitonic (i.e. sharp features at a well-defined energy, rather than broad edges), even at 300 K;
- the well can be used to hold electrons and holes in close proximity, to encourage more efficient recombination in e.g. lasers and LEDs.³

³See *Semiconductor physics*, by K. Seeger (Springer, Berlin, 1991) Section 13.2.

Applications include the Quantum-confined Stark effect modulator, which is dealt with in detail in e.g. *Low-dimensional semiconductor structures*, by M.J. Kelly (Clarendon Press, Oxford, 1995) Section 18.8.

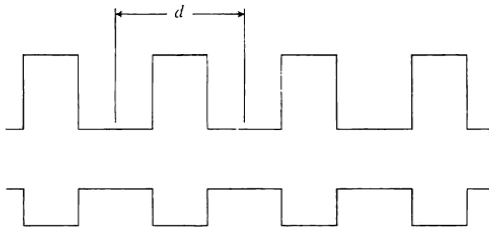


Fig. 7.7 Schematic of the band-edges in a superlattice.

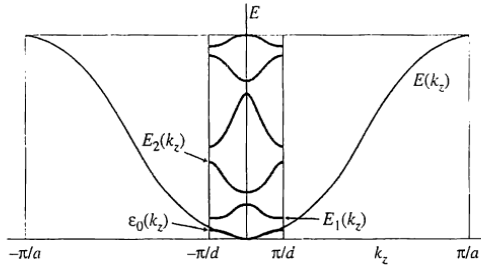


Fig. 7.8 Minibands and minigaps in a superlattice (see Fig. 7.7). Here, π/a denotes the Brillouin-zone boundary of the underlying lattice; $E(k_z)$ represents the unperturbed, original bulk band. The effect of the superlattice, of period d , is to introduce new 'mini Brillouin zones' with boundaries at $\pm\pi/d$. The band is split into minibands with dispersion relationships labelled by $E_0(k_z)$, $E_1(k_z)$, $E_2(k_z)$ etc.

7.3.6 Superlattices

A superlattice contains a set of quantum wells which are sufficiently closely spaced for the carriers to tunnel between wells (see Fig. 7.7). The situation is analogous to the tight-binding or Kronig–Penney models of bandstructure; the subbands will broaden out to form minibands with minigaps between. One can also think of the periodicity of the superlattice introducing a new set of Brillouin-zone boundaries and hence energy gaps (see Fig. 7.8).

Superlattices have a variety of applications in resonant tunnelling structures and far-infrared detectors; we shall discuss more of the details of these devices in Chapter 12.⁴

7.3.7 Type I and type II superlattices

Type I superlattices

Thus far, we have been using the system GaAs-(Ga,Al)As as an example in our discussion of quantum wells and superlattices.⁵ In GaAs-(Ga,Al)As superlattices, the band offsets of the two materials are such that the electron and hole wells occur in the same material (GaAs).⁶ Such an arrangement is known as a *Type I* superlattice; other examples include the systems (Ga,In)As-InP, (Ga,In)As-(Al,In)As and (Cd,Hg)Te-CdTe. Note that the band offsets vary between the different systems, so that it is possible to design superlattices in which, for example, the holes are strongly confined in deep wells, and the electrons are only weakly confined in very shallow wells.

Type II superlattices: uncrossed bands

In a Type II superlattice, the electron wells occur in one material and the hole wells occur in the other. Type II superlattices can be divided into two further

⁴See also See *Low-dimensional semiconductor structures*, by M.J. Kelly (Clarendon Press, Oxford, 1995) Chapters 9 and 10, 17–19.

⁵The various material systems mentioned in this Section are described in more detail in *Low-dimensional semiconductor structures*, by M.J. Kelly (Clarendon Press, Oxford, 1995).

⁶By implication, the *barriers* for electrons and holes are both provided by the (Ga,Al)As.

Fig. 7.9 Schematic band line-up in type II superlattices without crossed gaps (c.b. = conduction band-edge; v.b. = valence band-edge). Two examples are shown to illustrate the effect that varying the band offsets can have; (a) has a deep electron well in material B; (b) has a deep hole well in material B.

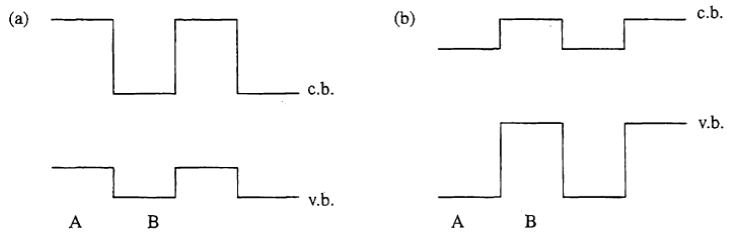
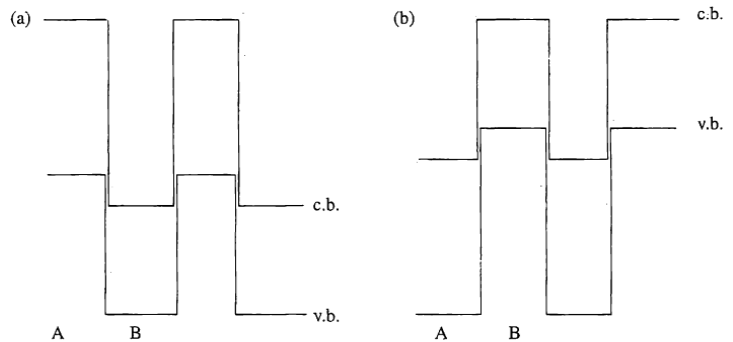


Fig. 7.10 Schematic band line-up in Type II superlattices with crossed gaps (c.b. = conduction band-edge; v.b. = valence band-edge). Two examples are shown to illustrate the effect that varying the band offsets can have; (a) is similar to the line-up in the InAs-(Ga,In)Sb system.



classes according to whether or not the electron wells are at a higher energy than the hole wells. The simplest situation occurs in superlattices where the bottom of the electron wells is at a higher energy than the top of the hole wells (the bands are said to be uncrossed). Fig. 7.9 shows two possible arrangements of the bands. An interesting aspect of such systems is that the hole and electron states are now somewhat spatially separated; in the case of wide layers, the electron subband wavefunctions will be mainly in one material and those of the holes in the other. In the case of superlattices, the electron and hole miniband wavefunctions will have greatest amplitude in different layers. This spatial separation will greatly modify the strength of the optical transitions between hole and electron states.

Examples of Type II superlattices without crossed gaps include Si-Ge and (Ga,In)As-GaAs.⁷

⁷ Note that it is in principle possible to make a Type II superlattice using two semiconductors with equal band gaps.

Type III superlattices: Type II superlattices with crossed gaps

In Type II systems with a crossed gap, the electron wells are at a lower energy than the hole wells (these are also known as 'Type III superlattices'). The situation is shown in Fig. 7.10; the band-gap of material B lies entirely outside the band gap of material A. The InAs-(Ga,In)Sb system is an example of a Type II crossed-gap superlattice. (Ga,In)Sb has the larger band gap, and its conduction band lies at a higher energy than that of InAs; the band line-up is similar to that in Fig. 7.10(a).

InAs-(Ga,In)Sb superlattices are of some interest because a semiconductor-to-semimetal transition can be engineered by varying the well widths. The energetic separation of the electron states (in the InAs) and the hole states

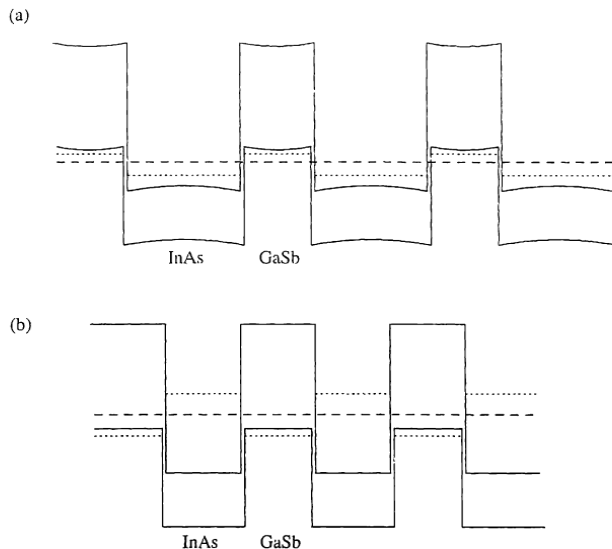


Fig. 7.11 Band profile of InAs-(Ga,In)Sb superlattices, showing the lowest electron and hole states in the InAs and (Ga,In)Sb respectively (dotted lines). The chemical potential is shown as a dashed line. (a) A semimetallic arrangement (wide InAs layers), in which transfer of electrons from (Ga,In)Sb to InAs occurs, leaving behind holes. (b) A semiconducting arrangement (narrow InAs layers) in which the lowest hole state (in (Ga,In)Sb) is lower than the lowest electron state (in InAs), with the chemical potential in between.

(in the (Ga,In)Sb) is mainly determined by the InAs well width, since the InAs conduction-band effective mass is much smaller than that of the holes in (Ga,In)Sb. If the layers are wide (see Fig. 7.11(a)), the electron states in InAs are below the hole states in (Ga,In)Sb; electron transfer occurs from the valence band of (Ga,In)Sb (leaving holes behind) to the conduction-band well of InAs. The system is thus an intrinsic semimetal;⁸ electrons and holes are simultaneously present without doping.

As the thickness of the layers decreases, the confinement energies increase until the electron states in InAs are above the hole states in (Ga,In)Sb (Fig. 7.11(b)), resulting in a semiconducting system. The cross-over occurs at an InAs layer thickness of about 85 Å. The semiconducting superlattices are of potential interest as far-infrared photodetectors.

7.3.8 Heterojunctions and modulation doping

A heterojunction is a single junction between two layers of different semiconductors. A typical example is the GaAs-(Ga,Al)As heterojunction. First, a thick layer of undoped GaAs is grown, followed by a few tens of nm of (Ga,Al)As (the *spacer layer*). After this a section of heavily-doped ($n \sim 10^{18} \text{ cm}^{-3}$) (Ga,Al)As is deposited, followed by more undoped (Ga,Al)As (not shown). The final band arrangement is shown in Fig. 7.12. The combination of the conduction band offset at the interface and the pinning of the chemical potential inside the GaAs and (Ga,Al)As layers results in a one-dimensional, approximately triangular potential well containing electrons which have 'dropped off' the donors in the (Ga,Al)As. As in the case of the quantum well, the electrons' *z*-direction motion is quantised into subbands; however, the *xy* motion is unconstrained, so that a degenerate (metallic) two-dimensional electron system results.

⁸ It has equal numbers of electrons and holes, and both are degenerate systems (i.e. they possess Fermi surfaces).

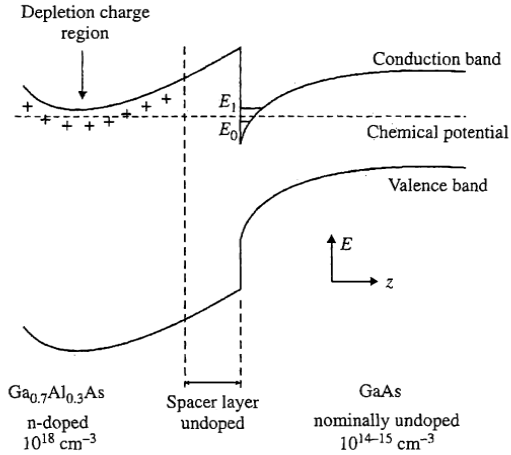


Fig. 7.12 The band-edges in a modulation-doped GaAs-(Ga,Al)As heterojunction.

The spacer layer separates the ionised donors from the electrons, dramatically reducing the scattering; low-temperature mobilities $\mu_c \sim 10^7 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ are possible using this technique, which is known as **modulation doping** (because the dopant concentration in the (Ga,Al)As is modulated).

Applications of heterojunctions include the **High-electron-mobility transistor (HEMT)**, a field-effect transistor with a heterojunction as active layer.⁹ Most experiments on the **quantum Hall effect** are carried out using heterojunctions; we shall study this effect in detail in Chapter 11.

Other applications of modulation doping include the fabrication of so-called δ -doping layers, in which a very heavy dose of dopant atoms is deposited in a small width ($\sim$ a few monolayers). In effect, this forms a two-dimensional impurity band, a very thin metallic layer within the host semiconductor. Such layers are finding applications as metallic gates and contacts *within* complex, layered semiconductor structures.

7.3.9 The envelope-function approximation

The envelope-function approximation is often used to calculate the electronic energy levels in heterostructures, such as GaAs-(Ga,Al)As quantum wells, which are made up from two or more similar semiconductors. Because the two semiconductors involved are very similar from both chemical and crystallographic points of view, it is assumed that the rapidly-oscillating part of the Bloch function (the part which has the periodicity of the lattice) is the same in both materials; only the *envelopes* ϕ of the Bloch functions differ in the two semiconductors.

As an example, consider the GaAs-(Ga,Al)As quantum well shown in Fig. 7.4; let a typical envelope function in the GaAs well be $\phi_A(z)$ and the corresponding envelope function in the (Ga,Al)As barriers be $\phi_B(z)$. The boundary conditions in the envelope function approximation at the interfaces (i.e. at $z = \pm \frac{a}{2}$) are

$$\phi_A = \phi_B \quad (7.2)$$

⁹See *Low-dimensional semiconductor structures*, by M.J. Kelly (Clarendon Press, Oxford, 1995) Chapter 16.

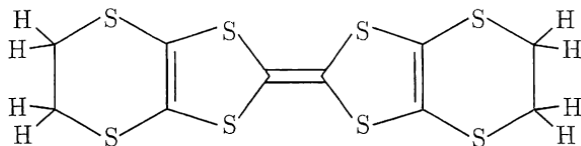


Fig. 7.13 The bisethylenedithiotetrathiofulvalene (BEDT-TTF) molecule.

and

$$\frac{1}{m_A^*} \frac{d\phi_A}{dz} = \frac{1}{m_B^*} \frac{d\phi_B}{dz}, \quad (7.3)$$

where m_A^* and m_B^* are the effective masses in the well and barrier respectively. Equation 7.3 ensures that the probability-density flux is conserved.

By considering envelope functions such as $\cos(kz)$ and $\sin(kz)$ in the well and evanescent waves $e^{\pm kz}$ in the barriers, the subband energies can be found. The exercises contain an example of this technique, which can easily be extended to more complex structures such as superlattices.

7.4 Band engineering using organic molecules

7.4.1 Introduction

In the earlier sections of this chapter, the problem of designing 'tailor-made' bandstructure was approached by using multilayers of different semiconductors to form superlattices and quantum wells. In this section, we shall examine another approach which involves the use of organic molecules to 'build' systems with easily controllable electronic properties, so-called organic molecular metals. The relationship between the crystal structures of these materials and their bandstructure may be easily understood using the tight-binding model, and so much of this section represents a useful revision of Chapter 4 and the other chapters on bandstructure.

Organic molecular metals are the subject of a considerable research effort world-wide; the techniques involving high magnetic fields which are described later in Chapter 8 are extensively applied in the study of the Fermi surfaces of organic molecular metals, and one of these materials will be used as a case-study in that chapter.

7.4.2 Molecular building blocks

In our discussion of the tight-binding model (Chapter 4), we saw that atomic orbitals were used to build up the Bloch functions which give rise to the bandstructure. Similarly, the bandstructure of organic molecular metals is derived from molecular orbitals. Bisethylenedithiotetrathiofulvalene (also known as BEDT-TTF or ET; Fig. 7.13) is typical of the molecules used as 'bandstructure building-blocks' in organic molecular metals.¹⁰ It is roughly flat, so that it can be stacked in a variety of arrangements in a solid, and it is surrounded by voluminous molecular orbitals. In order to form the bandstructure, BEDT-TTF molecules must then be stacked next to each other so that the molecular orbitals overlap; crudely one might say that this enables the electrons to transfer from molecule to molecule.

¹⁰Other molecules used for this purpose are described in *Organic superconductors* by T. Ishiguro, K. Yamaji and G. Saito (Springer-Verlag, Berlin 1998) and *Fermi surfaces of low-dimensional organic metals and superconductors* by J. Wosnitza (Springer-Verlag, Berlin 1996).

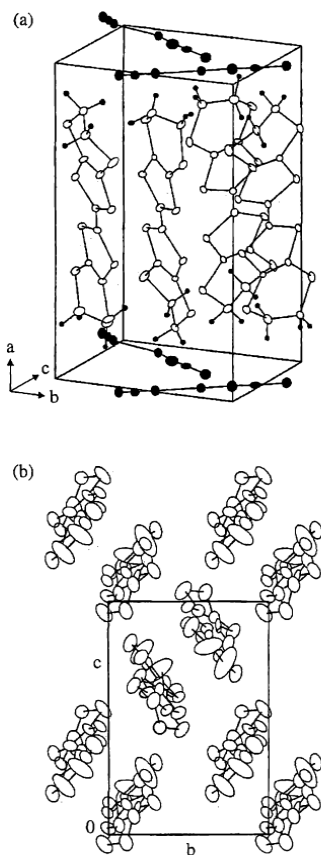


Fig. 7.14 Structure of the BEDT-TTF charge-transfer salt κ -(BEDT-TTF)₂Cu(NCS)₂. (a) Side view of molecular arrangement; the BEDT-TTF molecules pack in planes separated by layers of the smaller Cu(NCS)₂ anions. (b) View downwards onto the BEDT-TTF planes, showing that the BEDT-TTF molecules are packed closely together, allowing substantial overlap of the molecular orbitals. (Based on H. Urayama *et al.*, *Chem. Lett.* 1988, 463.)

The disposition of the bandstructure-forming molecules in an ordered arrangement is usually accomplished by making a charge-transfer salt. In a charge-transfer salt, a number j of BEDT-TTF molecules will jointly donate an electron to a second type of molecule (or collection of molecules) which we label X , to form the compound (BEDT-TTF) _{j} X ; owing to its negative charge, X is known as the anion. The transfer of charge serves to bind the charge-transfer salt together (in a manner analogous to ionic bonding) and also leaves behind a hole, jointly shared between the j BEDT-TTF molecules. This means that the bands formed by the overlap of the BEDT-TTF molecules will be partially filled; i.e. the resulting salt will be metallic.

Figure 7.14 shows the molecular arrangements in the BEDT-TTF charge-transfer salt κ -(BEDT-TTF)₂Cu(NCS)₂. (Here the κ denotes the packing arrangement of the BEDT-TTF molecules; often several different packing arrangements can be achieved with one particular anion.) The BEDT-TTF molecules are packed into layers, separated by layers of the Cu(NCS)₂ anion molecules. Within the BEDT-TTF layers, the molecules are in close proximity to each other, allowing substantial overlap of the molecular orbitals; in the language of Section 4.2.3, we can say that the transfer integrals for hopping of carriers between BEDT-TTF molecules will be relatively large within the BEDT-TTF planes. Conversely, in the direction perpendicular to the BEDT-TTF planes, the BEDT-TTF molecules are well separated from each other; the transfer integrals will be much smaller in this direction. This extreme anisotropy, resulting from a layered structure, is typical of most organic molecular metals; it results in bandstructures and electronic properties which for many purposes can be considered to be two-dimensional.

Figure 7.15 shows another BEDT-TTF charge-transfer salt, β -(BEDT-TTF)₂I₃ (again β denotes the packing arrangement of the BEDT-TTF molecules). As in the case of κ -(BEDT-TTF)₂Cu(NCS)₂ (Fig. 7.14), the BEDT-TTF molecules pack in planes separated by layers of the smaller anions, in this case I₃ molecules. However, the packing arrangement of the BEDT-TTF molecules is considerably different from that in κ -(BEDT-TTF)₂Cu(NCS)₂.

A comparison of Figs 7.14 and 7.15 shows that charge-transfer salts are an extremely flexible system with which to study the physics of band formation; by changing the anion, one can make the BEDT-TTF molecules pack in different arrangements. As the BEDT-TTF molecules are long and flat (unlike single atoms, which are far more like spheres), the transfer integrals will depend strongly on the way in which the BEDT-TTF molecules are arranged with respect to each other; this will be reflected in the *shape* of the resulting bands.

Far more subtle changes are also possible. β -(BEDT-TTF)₂I₃ is a member of the so-called β -phase BEDT-TTF salts, which have the generic formula β -(BEDT-TTF)₂ X , where the anion molecule X can be I₃, IBr₂, AuI₂ etc. The different possible anions have slightly different lengths; therefore the size of the unit cell can be varied by using different anions (see Fig. 7.15). If a longer anion (e.g. $X = \text{I}_3$) is used, the BEDT-TTF molecules will be further apart, so that the transfer integrals will be smaller; the resulting bands will be narrower and the effective masses larger. Conversely, if a shorter anion (e.g. $X = \text{IBr}_2$) is used, the BEDT-TTF molecules will be closer, so that the transfer integrals will be larger; the resulting bands will be wider and the effective masses smaller.

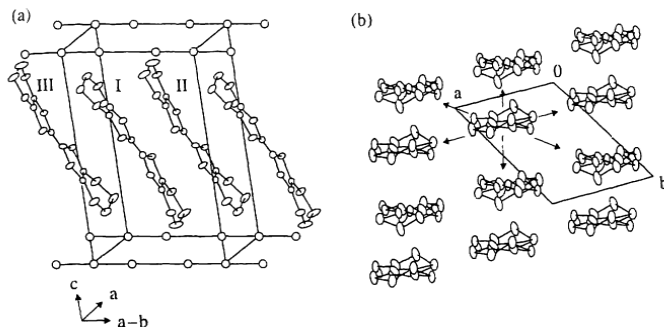


Fig. 7.15 Structure of the BEDT-TTF charge-transfer salt β -(BEDT-TTF) $_2$ I $_3$. (a) Side view of molecular arrangement. As in Fig. 7.14, the BEDT-TTF molecules pack in planes separated by layers of the smaller anions; in this case the anions are I $_3$ molecules. (b) View downwards onto the BEDT-TTF planes, showing that the BEDT-TTF molecules are packed closely together, allowing substantial overlap of the molecular orbitals. However, the packing arrangement of the BEDT-TTF molecules is considerably different from that in κ -(BEDT-TTF) $_2$ Cu(NCS) $_2$ (see Fig. 7.14). (Based on H. Kuroda *et al.*, *Synth. Met.* **27**, A491 (1988).)

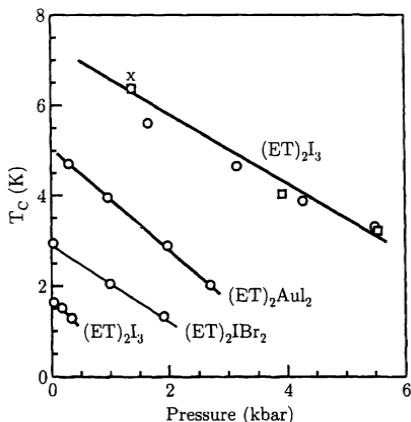


Fig. 7.16 Superconducting critical temperature T_c versus hydrostatic pressure for three β -phase BEDT-TTF salts (BEDT-TTF has been abbreviated to ET to save space). Decreasing the unit cell size, either by using a shorter anion or by increasing the pressure, reduces T_c . (Note that β -(BEDT-TTF) $_2$ I $_3$ undergoes a structural phase transition at about 0.6 kbar.) (Based on data compiled in *Organic Superconductors*, by T. Ishiguro *et al.* (Springer, 1998).)

Thus, different anions can be employed to optimise a particular physical property; an example of this is shown in Fig. 7.16, which shows the relationship between superconducting critical temperature T_c (see Appendix H) and hydrostatic pressure for three β -phase salts. It can be seen that increasing the pressure (which decreases the unit cell size, thereby increasing the width of the bands) lowers T_c ; substitution of a larger anion increases the unit cell size, countering the effect. We shall see why the bandwidth affects T_c in Chapter 8.

7.4.3 Typical Fermi surfaces

A section through the first Brillouin zone and Fermi surface of κ -(BEDT-TTF) $_2$ Cu(NCS) $_2$ is shown in Fig. 7.17. The section shown there is parallel to the highly-conducting planes; as has already been remarked, the electronic properties of the material are largely two-dimensional. The Fermi surface may be understood by examining Fig. 7.14(b). Note that the BEDT-TTF molecules are packed in pairs called dimers; if we consider each dimer as a unit, it will be seen that it is surrounded by four other dimers. We thus expect the bandstructure to be fairly isotropic in the plane, because there will be substantial transfer integrals in these four directions. The unit cell contains two

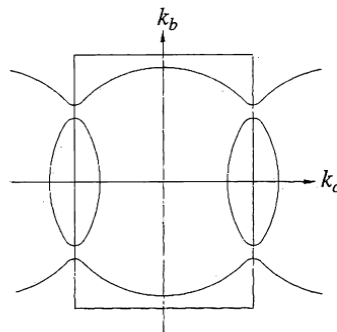


Fig. 7.17 Brillouin zone and Fermi surface of κ -(BEDT-TTF) $_2$ Cu(NCS) $_2$, showing the open, quasi-one-dimensional sections, and the closed, quasi-two-dimensional pocket. (J.M. Caulfield *et al.*, *J. Phys.: Condens. Matter*, **6**, 2911 (1994).)

dimers, each of which contributes a hole (in the charge-transfer process, two BEDT-TTF molecules jointly donate one electron to the anion). Thus, to a first approximation, the Fermi surface for holes might be expected to be roughly circular with the same area as the Brillouin zone (as there are two holes per unit cell). As can be seen from Fig. 7.17, the Fermi surface is *roughly* like this; however, the Fermi surface intersects the Brillouin zone boundaries in the c direction, so that band gaps open up. The Fermi surface thus splits into open (electron-like) sections running down two of the Brillouin-zone edges and a closed hole pocket straddling the other.

7.4.4 A note on the effective dimensionality of Fermi-surface sections

The Fermi surface of κ -(BEDT-TTF) $_2$ Cu(NCS) $_2$ has been shown to consist of both open and closed sections (see Fig. 7.17); it is customary to label such sections ‘quasi-one-dimensional’ and ‘quasi-two-dimensional’ respectively. The names arise because the group velocity $\mathbf{v}$ of the electrons is given by eqn 5.5

$$\hbar \mathbf{v} = \nabla_{\mathbf{k}} E(\mathbf{k}).$$

The Fermi surface is a surface of constant energy; eqn 5.5 shows that the velocities of electrons at the Fermi surface will be directed perpendicular to it. Therefore, referring to Fig. 7.17, electrons on the closed Fermi-surface pocket can possess velocities which point in any direction in the (k_b, k_c) plane; they have freedom of movement in two dimensions and are said to be **quasi-two-dimensional**. By contrast, electrons on the open sections have velocities predominantly directed parallel to k_b and are **quasi-one-dimensional**.

Quasi-two-dimensional and quasi-one-dimensional Fermi-surface sections are observed in many layered materials, and so it is worth getting to know how to identify them.

7.5 Layered conducting oxides

Oxides of transition metal ions such as Mn, Fe, Ni, V, Ru and Cu have received considerable attention because of their electronic properties. In this Section I

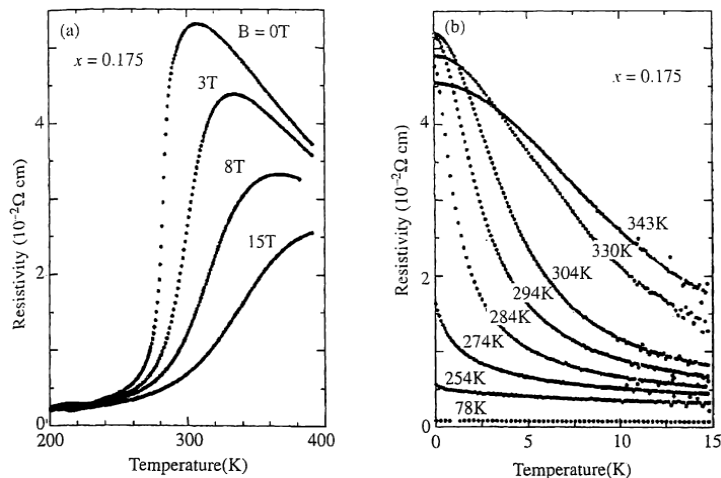


Fig. 7.18 (a) The resistivity of $\text{La}_{1-x}\text{Sr}_x\text{MnO}_3$ ($x = 0.175$) as a function of temperature for various different magnetic fields. (b) The resistivity of $\text{La}_{1-x}\text{Sr}_x\text{MnO}_3$ ($x = 0.175$) as a function of field for various different temperatures. (After Y. Tokura *et al.*, *J. Phys. Soc. Jpn.* 63, 3931 (1994).)

shall describe the oxides of Mn as an example of this interest, and show how the introduction of layered crystal structures can be used in attempts to optimise desirable properties.

Figure 7.18 shows the electrical resistivity of $\text{La}_{1-x}\text{Sr}_x\text{MnO}_3$ ($x = 0.175$) as a function of temperature for various different magnetic fields. At first, it might seem surprising that such a compound should conduct electricity; however, this occurs because the Mn ion can exhibit what is called mixed valence. In other words, the Mn ion can exist as *either* Mn^{3+} or Mn^{4+} . The parent compound, LaMnO_3 , is an insulator in which all of the Mn ions are in the Mn^{3+} state (La is a trivalent ion, La^{3+}). In $\text{La}_{1-x}\text{Sr}_x\text{MnO}_3$, trivalent La ions are replaced by divalent Sr^{2+} ions, resulting in a shortage of electrons (or a surplus of holes); hence a fraction x of the Mn ions will lose an extra electron, so that they will end up in the Mn^{4+} state.

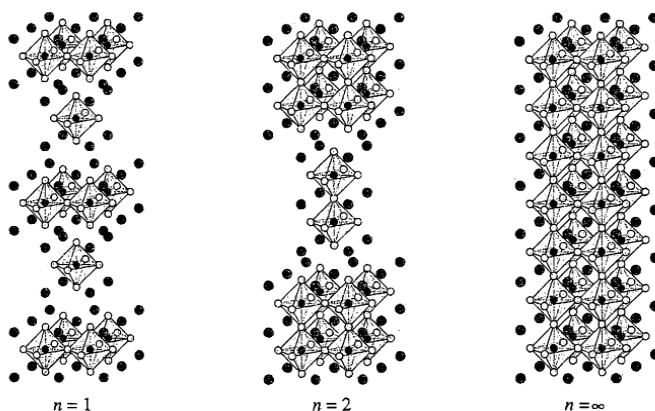
The important region of the bandstructure for our purposes involves two quantum levels associated with the Mn ions and their surrounding O ions. The lower level is highly localised on the Mn ions, and is known as the t_{2g} level. Exchange interactions¹¹ dictate that three spin-polarised electrons can be accommodated in this level; the subsequent electron should be placed in the level above (the e_g level), and should also have the same spin polarisation as the t_{2g} electrons. Thus, Mn^{3+} ions have three electrons in the t_{2g} level and one electron in the e_g level, whereas the Mn^{4+} ions have empty e_g levels.

Whereas the t_{2g} levels are very tightly bound (effectively rendering the electrons therein immobile), the e_g levels overlap to form a narrow band, via which electrons can hop from Mn to Mn; oxides in which there is a mixture of Mn^{3+} and Mn^{4+} therefore have a partly filled band of mobile electrons which can lead to electrical conduction. The spectacular dependence of the resistivity of $\text{La}_{1-x}\text{Sr}_x\text{MnO}_3$ on magnetic field and temperature occurs because of another manifestation of the exchange interactions, known as double exchange,¹² which strongly suppresses hopping of an e_g electron from an Mn^{3+} ion to an Mn^{4+} ion *unless* both have their t_{2g} spins polarised in

¹¹For the present purposes, these interactions should just be accepted as fact, as this book does not seek to present a detailed description of magnetism in solids. For a full discussion of exchange interactions in solids, see *Magnetism in condensed matter*, by S. Blundell (OUP, 2001).

¹²See Chapter 4 of *Magnetism in condensed matter*, by S. Blundell (OUP, 2001).

Fig. 7.19 Ruddlesden–Popper oxides (general formula $A_{n+1}B_nO_{3n+1}$) with $n = 1, 2, \infty$. The parameter n represents the number of layers of vertex shared BO_6 octahedra stacked along the crystallographic direction [001]. Here the A ions are grey, the B ions black and the O ions white. (Figure courtesy of Steve Blundell, Oxford University).



the same direction. $La_{1-x}Sr_xMnO_3$ ($x = 0.175$) undergoes a transition into a ferromagnetic state in which all of the t_{2g} electron spins are aligned at about 300 K; hence, the e_g electrons will be able to hop with ease and a relatively low resistance will result; this is the explanation of the large drop in resistivity at around 300 K in Fig. 7.18.

At temperatures above 300 K, hopping of the e_g electrons will be suppressed, as the spin system is unpolarised. However, the application of a magnetic field will act to align the spins of all of the electrons on the Mn ions, so that hopping will become easier, leading to the spectacular decreases in resistivity seen in Fig. 7.18. Such very large decreases in resistivity caused by the application of a magnetic field have become known as **colossal magnetoresistance**,¹³ materials exhibiting colossal magnetoresistance have great potential as, for example, resistive (as opposed to inductive) read heads for magnetic recording media.

The performance of oxides of Mn exhibiting colossal magnetoresistance depends on the interactions which cause the magnetic order, and the overlap of the Mn ions with their neighbours which determines the properties of the e_g band. One way of modifying these effects involves changing the effective dimensionality of these materials. $La_{1-x}Sr_xMnO_3$ is a member of a series of compounds known as Ruddlesden–Popper oxides, which possess a general formula $A_{n+1}B_nO_{3n+1}$; here n represents the number of layers of vertex shared BO_6 octahedra stacked along the crystallographic direction [001]. Examples with $n = 1$ (A_2BO_4), 2 ($A_3B_2O_7$) and ∞ (ABO_3) are shown in Fig. 7.19; it will be seen that the structures with $n = 1, 2$ are quasi-two-dimensional, in that the BO_6 octahedra which are responsible for the e_g band are planes which are rather well separated in the [001] direction. By contrast, $La_{1-x}Sr_xMnO_3$ corresponds to the case $n = \infty$, and possesses much more isotropic (i.e. more ‘three-dimensional’) bandstructure and magnetic interactions.¹⁴

Much work continues on the effect of varying the effective dimensionality of oxide materials in an attempt to optimise desirable properties. Amongst studies of this kind which have yielded great gains is the development of the ‘high T_c ’ cuprate superconductors. Values of T_c well in excess of 100 K have been

¹³ A much smaller magnetoresistance is seen in very many electrically conductive materials, but its origin is rather different; it is usually caused by modification of the real-space paths of the carriers due to the Lorentz force. This form of magnetoresistance will be dealt with in Chapter 10.

¹⁴ The $n = \infty$ case is often referred to as the perovskite structure.

achieved through the production of structures in which the electronically active layers containing Cu and O have been progressively separated by including more intermediate layers of 'packing'. A detailed discussion of this is beyond the scope of this book, but an excellent summary of the development of layered cuprate superconductors is given in Chapter 7 of *Superconductivity*, by C.P. Poole, H.A. Farach and R.J. Creswick (Academic Press, San Diego, 1995) and in *Superconductivity*, by J. Annett (OUP, 2002).

7.6 The Peierls transition

Many of the layered substances (organic molecular solids, oxides) discussed in this chapter have Fermi-surface sections which are quasi-one-dimensional (see Section 7.4.4 for an explanation of this terminology). Such substances are unstable with respect to what are known as **Peierls transitions**, in which a periodic modulation of the crystal structure results. Figure 7.20 describes

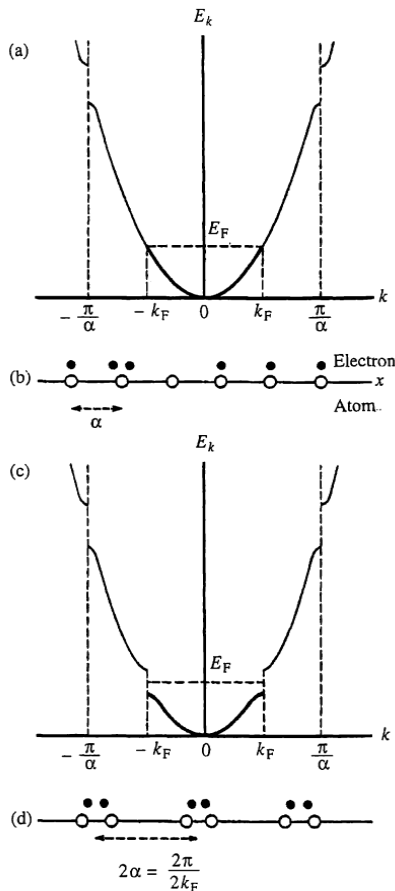


Fig. 7.20 (a) A simple nearly-free-electron band in a monovalent one-dimensional material; band gaps exist at the Brillouin-zone boundary and the Fermi surface (a pair of points) has a width equal to half that of the Brillouin zone. (b) The equivalent of (a) plotted in r -space, showing (on average) one electron (black dot) per atom (hollow point); an electron has been 'caught in the act' of hopping from one atom to the next. (c) The band after the Peierls distortion shown in (d). The crystal has distorted, doubling the r -space unit-cell size, and halving the Brillouin-zone size. As new zone boundaries exist at $\pm k_F$, a band-gap opens up there, lowering the electronic energy, and turning the substance into an insulator, with two electrons per unit cell.

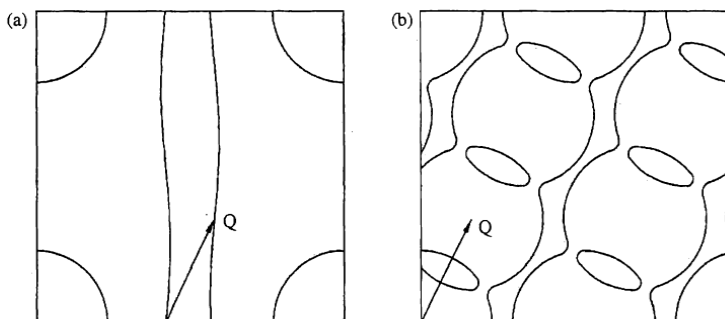


Fig. 7.21 (a) A Fermi surface (consisting of a pair of quasi-one-dimensional sheets plus a quasi-two-dimensional cylinder) which is prone to spin- or charge-density wave formation. The nesting vector Q which maps the quasi-one-dimensional Fermi-surface sections onto each other (so that a band-gap opens along their whole length) is shown as an arrow. (b) Rearrangement of the Fermi surface in (a) caused by a spin- or charge-density wave (the arrow indicates the nesting vector). A new k -space periodicity and new Brillouin zone have been introduced; the remaining section of Fermi surface has been 'cut and pasted' with this periodicity, resulting in a much altered bandstructure.

this schematically in a one-dimensional material. Figure 7.20(a) displays a nearly-free-electron band in a monovalent one-dimensional material. As we saw in Chapter 3, band gaps exist at the Brillouin-zone boundary at $\pm \frac{\pi}{a}$ and the Fermi surface (a pair of points) is half the length of the Brillouin zone (i.e. $k_F = \pm \frac{\pi}{2a}$). The situation in r -space is shown in Fig. 7.20(b); on average there is one electron per atom, and electrons are free to move along the chain of atoms, which are spaced by a . It turns out that in this situation, a very simple distortion of the r -space lattice can lower the energy of the electron system. Such a distortion is shown in Fig. 7.20(d); adjacent atoms have moved together to form dimers; the unit-cell length has doubled to $2a$, so that the Brillouin zone has halved in size (see Fig. 7.20(c); the new boundaries are at $\pm k_F = \pm \frac{\pi}{2a}$). A band-gap opens up at the new Brillouin-zone boundaries, lowering the electronic energy, and turning the substance into an insulator. This sort of distortion will be able to occur as long as the electrons can interact with the lattice.

The distortion has resulted in electronic charge piling up on the dimers; the resulting state is known as a **charge-density wave**, as the electron density (once much more uniform) has become modulated. A similar lowering of energy can be obtained by doubling the unit cell with a periodic modulation of the electron spin; this is known as a **spin-density wave**.¹⁵

Modulations of this type may be described by a series of terms $\propto \cos(Qx)$, where the wavevector $Q (= 2k_F)$ is known in the trade as a **nesting vector**.

It turns out that charge- and spin-density waves are likely to occur in non-one-dimensional materials whenever a single nesting vector can cause large areas of Fermi surface to map onto each other; in r -space terms, this means introducing a distortion with a periodicity which creates band gaps over large sections of Fermi surface, so that a large fraction of the electrons can lower their energy. Fig. 7.21 illustrates such a case.

¹⁵ See Chapter 8 of *Magnetism in condensed matter*, by S. Blundell (OUP, 2001) and Chapter 2 of *One-dimensional conductors*, by S. Kagoshima, H. Nagasawa and T. Sambongi (Springer-Verlag, Berlin, 1988).

Peierls transitions are remarkably common in organic molecular metals and oxides and are even observed in a few elemental metals such as chromium.

Further reading

A more detailed treatment of semiconductor heterostructures is found in *Quantum semiconductor structures* by Claude Weisbuch and Borge Winter (Academic Press, San Diego, 1991) and in *Low-dimensional semiconductor structures*, by M.J. Kelly (Clarendon Press, Oxford, 1995), Chapters 2, 9 and 10; a complementary (simple) summary is given in *Solid state physics*, by G. Burns (Academic Press, Boston, 1995), Sections 18.1–18.6. The book *Quantum wells, wires and dots: theoretical and computational physics*, by Paul Harrison (Wiley, Chichester, 1999) is an excellent point of departure if you want to carry out calculations and simulations. An good review of the envelope-function model is given in *Structure de bandes, niveaux d'impureté et effets Stark dans les super-réseaux* by G. Bastard, *Acta Electronica* **25**, 147–161 (1983). Optical properties of semiconductor structures are tackled in depth in *Optical properties of solids*, by M. Fox (OUP, 2001).

Reviews of organic molecular metals are given in *Organic superconductors* by T. Ishiguro, K. Yamaji and G. Saito (Springer-Verlag, Berlin 1998), *Fermi surfaces of low-dimensional organic metals and superconductors* by J. Wosnitza (Springer-Verlag, Berlin 1996) and by J. Singleton in *Reports on Progress in Physics* **63**, 1111 (2000).

Electrically conductive oxides are discussed in *Transition metal oxides*, by P.A. Cox (OUP, 1995) and in *Metal-insulator transitions*, by N.F. Mott (Taylor and Francis, London, 1990 (second edition)). Relatively recent reviews of colossal magnetoresistance have been provided by A.P. Ramirez (*J. Phys.: Condens. Matter* **9**, 8171 (1997)) and C.N.R. Rao (*J. Mater. Chem.* **9**, 1 (1999)).

An excellent review of Peierls transitions (at a reasonably simple level) is given in Chapter 2 of *One-dimensional conductors*, by S. Kagoshima, H. Nagasawa and T. Sambongi (Springer-Verlag, Berlin, 1988). See also G. Grüner, *Rev. Mod. Phys.* **60**, 1129 (1988) and *Rev. Mod. Phys.* **66**, 1 (1994).

Exercises

The problems in this chapter explore the envelope-function approximation, using the GaAs-(Ga,Al)As quantum well shown in Fig. 7.22 as an example. The GaAs layer forms a one-dimensional potential well, sandwiched between $\text{Ga}_{0.65}\text{Al}_{0.35}\text{As}$ barriers. The energy gaps of GaAs and $\text{Ga}_{0.65}\text{Al}_{0.35}\text{As}$ are $E_{g1} = 1.52$ eV and $E_{g2} = 1.97$ eV respectively and you can assume that $\frac{\Delta E_c}{\Delta E_v} \approx 1.5$. The GaAs layer is 100 Å thick. Effective masses in GaAs and (Ga,Al)As are given in Table 7.1.

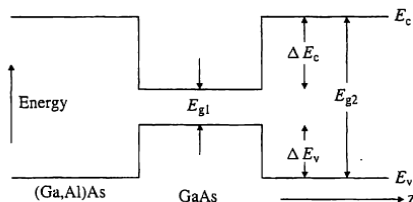


Fig. 7.22 Schematic of a GaAs-(Ga,Al)As quantum well.

Table 7.1 Effective masses (in units m_e) of GaAs and $\text{Ga}_{0.65}\text{Al}_{0.35}\text{As}$.

Material	Electron	Light hole	Heavy hole
GaAs	0.07	0.085	0.45
(Ga,Al)As	0.091	0.11	0.5

- (7.1) Let a typical envelope function in the GaAs well be $\phi_A(z)$ and the corresponding envelope function in the (Ga,Al)As barriers be $\phi_B(z)$. The boundary conditions in the envelope-function approximation at the interfaces (i.e. at $z = \pm \frac{a}{2}$) are

$$\phi_A = \phi_B \quad (7.4)$$

and

$$\frac{1}{m_A^*} \frac{d\phi_A}{dz} = \frac{1}{m_B^*} \frac{d\phi_B}{dz}, \quad (7.5)$$

where m_A^* and m_B^* are the effective masses in the well and barrier respectively. Explain why envelope functions such as $\cos(kz)$ and $\sin(kz)$ are appropriate for the well and $e^{\pm\kappa z}$ for the barriers, and show that the bound states

in the well are given by solutions of the equations

$$\tan\left(\frac{ka}{2}\right) = -\frac{m_B^* \kappa}{m_A^* k} \quad (7.6)$$

and

$$\tan\left(\frac{ka}{2}\right) = +\frac{m_A^* \kappa}{m_B^* k}. \quad (7.7)$$

- (7.2) Using eqns 7.6 and 7.7, and the well parameters given above, find the energies of the bound states for the electrons, heavy holes and light holes. (IMPORTANT HINT: Equations 7.6 and 7.7 cannot be solved analytically. The problem can either be done *numerically*, by writing a simple program in C, PASCAL or some other language or *graphically*, using your favourite spreadsheet.)
- (7.3) Label the state closest to the bottom of the well $i = 1$ and the next $i = 2$ etc. for each type of carrier (remember that the 'lowest' hole state is that closest to the GaAs valence band-edge). Sketch the low-temperature optical transmission of the well as a function of frequency ν in the range $E_{g1} \leq h\nu \leq E_{g2}$, assuming that the dominant transitions have $\Delta i = 0$.